

THE WEST AFRICAN EXAMINATIONS COUNCIL

BASIC EDUCATION CERTIFICATE EXAMINATION -- JUNE 2008

MATHEMATICS -- PAPER 1

Duration: 1 Hour | Level: JHS 3 / BECE | Code: MATH 01/02 - P1

INSTRUCTIONS TO CANDIDATES

Answer all 40 questions. Each question is followed by four options lettered A to D. Find the correct option for each question and mark it on your answer sheet. Mathematical tables and graph papers are provided where required.

PART I: OBJECTIVE QUESTIONS

1. $M = \{1, 2, 3, 4, 5, \dots, 20\}$, $Q = \{3, 4, 5, 6, 7, 8\}$ and $R = \{2, 3, 5, 7\}$. If Q and R are subsets of M , find $Q \cap R$.
 - A. $\{3, 5\}$
 - B. $\{5, 7\}$
 - C. $\{3, 5, 7\}$
 - D. $\{2, 3, 5, 7\}$
2. List the members of the set $\{x : 2 \leq x \leq 5\}$.
 - A. $\{2, 5\}$
 - B. $\{2, 3, 4\}$
 - C. $\{2, 3, 5\}$
 - D. $\{2, 3, 4, 5\}$
3. Round 8,921,465 to the nearest hundred.
 - A. 8,921,000
 - B. 8,921,400
 - C. 8,921,460
 - D. 8,921,500
4. Write 98 as a product of its prime factors.
 - A. 2×7
 - B. $2^2 \times 7$
 - C. 2×7^2
 - D. $2^2 \times 7^2$
5. Evaluate: $4(8 - 2) + 5(3 - 8)$.
 - A. -31
 - B. -1
 - C. 37
 - D. 49
6. Arrange the following in descending order of magnitude: 0.32, $\frac{2}{5}$, 27%, $\frac{1}{3}$.
 - A. 0.32, $\frac{2}{5}$, 27%, $\frac{1}{3}$
 - B. 0.32, $\frac{1}{3}$, $\frac{2}{5}$, 27%
 - C. $\frac{2}{5}$, $\frac{1}{3}$, 0.32, 27%
 - D. 27%, 0.32, $\frac{1}{3}$, $\frac{2}{5}$
7. The ratio 8:12 is equivalent to $y:9$. What is the value of y ?
 - A. 4
 - B. 5
 - C. 6
 - D. 7
8. Write 0.55 as a fraction in its lowest term.
 - A. $\frac{5}{10}$
 - B. $\frac{11}{20}$
 - C. $\frac{11}{50}$
 - D. $\frac{55}{100}$

9. Which of the following geometric solids can be formed from a net with a square base and four congruent triangular faces?

- A. Cube
- B. Square pyramid
- C. Cone
- D. Triangular prism

10. Simplify: $5w + 7p^2 - 4w + 3p^2$.

- A. $w + 4p^2$
- B. $w + 10p^2$
- C. $9w + 10p^2$
- D. $w - 10p^2$

11. Osei bought a hat for GHS 5.00. He sold it to Yaovi at a profit of 20%. How much did Yaovi pay for the hat?

- A. GHS 1.00
- B. GHS 4.00
- C. GHS 6.00
- D. GHS 7.00

12. Simplify: $2^9 / 2^3$.

- A. 2^3
- B. 2^6
- C. 2^{12}
- D. 2^{27}

13. Find the image of -3 under the mapping $x \rightarrow 2(x + 3)$.

- A. 0
- B. 6
- C. 12
- D. -6

14. A football field is 120 m long and 75 m wide. What is the perimeter of the field?

- A. 195 m
- B. 240 m
- C. 315 m
- D. 390 m

15. A tank holds 240 litres of water. How much water is in the tank when it is $\frac{4}{5}$ full?

- A. 48 litres
- B. 120 litres
- C. 192 litres
- D. 200 litres

16. A farmer has $6x$ sheep and $5y$ goats. He sells $3x$ sheep and $2y$ goats. How many animals are left after the sales?

- A. $3x - 3y$
- B. $3x + 3y$
- C. $9x + 7y$
- D. $3x + 7y$

17. The average rainfall for May was 140 mm, June was 210 mm, and July was 180 mm. What was the total amount of rainfall in May, June and July?

- A. 350 mm
- B. 420 mm
- C. 530 mm
- D. 560 mm

18. If $21_{\text{base } x} = 9_{\text{base ten}}$, find the base x .

- A. 3
- B. 4
- C. 5
- D. 6

19. Find the truth set of the inequality: $2x - 3 < 7$.

- A. $\{x : x < 5\}$
- B. $\{x : x > 5\}$
- C. $\{x : x < 2\}$
- D. $\{x : x > 2\}$

20. A car uses 5 litres of petrol for a journey of 60 km. How many litres will it use for a journey of 180 km at the same rate?
- 10 litres
 - 12 litres
 - 15 litres
 - 20 litres
21. If set $P = \{1, 2, 3, 4, 6, 12\}$ and set $Q = \{2, 3, 5, 7, 11\}$, find $P \cap Q$.
- $\{2, 3\}$
 - $\{1, 2, 3\}$
 - $\{2, 3, 5\}$
 - $\{1, 2, 3, 4, 6, 12\}$
22. Find the Highest Common Factor (HCF) of 18, 24 and 36.
- 3
 - 6
 - 12
 - 72
23. Evaluate: $\frac{3}{4} - (\frac{1}{2} + \frac{1}{8})$.
- $\frac{1}{8}$
 - $\frac{1}{4}$
 - $\frac{3}{8}$
 - $\frac{1}{2}$
24. Solve for x : $3(x - 2) = 2(x + 4)$.
- 10
 - 12
 - 14
 - 16
25. Calculate the simple interest on GHS 12,000 for 3 years at 5% per annum.
- GHS 1,200
 - GHS 1,500
 - GHS 1,800
 - GHS 2,400
26. The interior angle of a regular polygon is 120 deg. How many sides does it have?
- 5
 - 6
 - 7
 - 8
27. Convert 11011_{two} to a base ten number.
- 23
 - 27
 - 29
 - 31
28. If $x : y = 2 : 3$ and $y : z = 4 : 5$, find $x : z$.
- 8 : 15
 - 2 : 5
 - 6 : 10
 - 8 : 12
29. Factorize completely: $4a^2 - 9b^2$.
- $(2a - 3b)(2a - 3b)$
 - $(2a + 3b)(2a - 3b)$
 - $(4a - 9b)(a + b)$
 - $(2a + 9b)(2a - b)$
30. A bag contains 5 red balls and 7 blue balls. What is the probability of picking a red ball at random?
- $\frac{5}{12}$
 - $\frac{7}{12}$
 - $\frac{5}{7}$
 - $\frac{1}{5}$

31. Find the circumference of a circle whose diameter is 14 cm. [Take $\pi = \frac{22}{7}$]
- 22 cm
 - 44 cm
 - 88 cm
 - 154 cm
32. If vector $u = (3, -2)$ and vector $v = (-1, 5)$, find $u + 2v$.
- (1, 8)
 - (1, 3)
 - (2, 8)
 - (5, -4)
33. Find the gradient of the line passing through points A(2, 3) and B(4, 7).
- $\frac{1}{2}$
 - 1
 - 2
 - 4
34. The perimeter of a rectangle is 28 cm. If its length is 8 cm, find its width.
- 6 cm
 - 8 cm
 - 10 cm
 - 12 cm
35. Find the value of y if $2^y = 32$.
- 4
 - 5
 - 6
 - 7
36. Calculate the area of a right-angled triangle with base 6 cm and hypotenuse 10 cm.
- 24 cm^2
 - 30 cm^2
 - 48 cm^2
 - 60 cm^2
37. Find the volume of a cylinder with radius 7 cm and height 10 cm. [Take $\pi = \frac{22}{7}$]
- 770 cm^3
 - 1540 cm^3
 - 2200 cm^3
 - 3080 cm^3
38. Make t the subject of the formula: $v = u + at$.
- $t = (v - u) / a$
 - $t = (v + u) / a$
 - $t = v / (u + a)$
 - $t = a(v - u)$
39. If $3x + 2y = 12$ and $x = 2$, find the value of y .
- 2
 - 3
 - 4
 - 6
40. Find the bearing of town P from Q if the bearing of Q from P is 065 deg.
- 115 deg
 - 205 deg
 - 245 deg
 - 335 deg

OFFICIAL WAEC ANSWER KEY & SCORING GUIDE

For candidate self-assessment and teacher evaluation:

1. [A] 2. [D] 3. [D] 4. [C] 5. [B]
 6. [C] 7. [C] 8. [B] 9. [B] 10. [B]
 11. [C] 12. [B] 13. [A] 14. [D] 15. [C]

16. [B] 17. [C] 18. [B] 19. [A] 20. [C]
21. [A] 22. [B] 23. [A] 24. [C] 25. [C]
26. [B] 27. [B] 28. [A] 29. [B] 30. [A]
31. [B] 32. [A] 33. [C] 34. [A] 35. [B]
36. [A] 37. [B] 38. [A] 39. [B] 40. [C]